

On the Limits of Sufficient Statistics for Macroeconomic Policy Evaluation

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Abstract

We study when optimal monetary policy can be evaluated from two sufficient statistics—forecasts of policy objectives and impulse responses, as in Barnichon and Mesters [2023]—and when it cannot. We identify three economically important scenarios where the correction step needs information beyond the two statistics: (1) when expectation formation is endogenous to the policy regime (a Lucas critique); (2) under multiple rational-expectations equilibria; and (3) at binding constraints such as the zero lower bound. In a calibration, the misspecified correction reverses the sign of the recommended policy and yields welfare losses exceeding the correctable gain. We propose regime-robust extensions, with implications for credibility.

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1 Introduction

How much must a policymaker know about the structure of the economy to judge whether a policy is optimal—and to improve it if it is not? An influential research program gives a reassuring answer: far less than a fully specified structural model would seem to require. The *sufficient statistics* approach, developed in public finance by Chetty [2009], shows that the welfare effects of a policy change can often be written in terms of a few high-level, estimable objects—reduced-form elasticities—without recovering the deep structural parameters that generate them. Its appeal rests on two properties. First, a single welfare formula holds across the many structural models that share those elasticities, insulating the analyst from primitives that are hard to identify. Second, the required statistics map directly onto magnitudes that modern quasi-experimental methods estimate credibly. By building this bridge between structural and reduced-form analysis, the approach has reshaped applied welfare economics precisely because it appears to sidestep the structural estimation that policy evaluation was once thought to demand.

The force of the word *sufficient*, however, is always relative to a maintained class of models and perturbations. A statistic that summarizes welfare under one set of assumptions need not do so under another, and the elasticities themselves may move with the policy being evaluated. In the static, partial-equilibrium settings where the approach originated, the conditions for sufficiency are by now well understood. They are far less transparent in macroeconomics, where policy unfolds over time, agents are forward-looking, outcomes are pinned down in general equilibrium, or multiple equilibria, and the very mapping from policy to outcomes can depend on the policy regime in place. Whether—and when—a handful of sufficient statistics can support optimal macroeconomic policy evaluation is therefore an open and consequential question. It is the question this paper takes up.

Perhaps the most influential affirmative answer is due to Barnichon and Mesters [2023], henceforth BM, who show that two statistics already central to the practice of policymaking suffice to determine optimality: (i) conditional forecasts $\mathbb{E}_t \mathbf{Y}_0$ of the policy objectives, and (ii) the impulse responses R of those objectives to policy shocks. Here the impulse responses play the role of Chetty’s elasticities and the forecast summarizes the current state. The elegance of the result lies in its claimed invariance: the sufficiency of the two statistics holds, BM argue, whether the policymaker operates under commitment or discretion, and regardless of the mechanism through which private agents form expectations.

The central theoretical result is as follows. For a quadratic loss function and a linear model, BM show that the gradient of the loss with respect to policy shocks equals $R'W \mathbb{E}_t \mathbf{Y}_0$. Because a convex optimization problem requires a zero gradient at its solution, optimal policy is characterized by the condition $R'W \mathbb{E}_t \mathbf{Y}_0 = 0$. The *Optimal Policy Perturbation* (OPP)

is then defined as the closed-form correction

$$\hat{\delta}^* = - \left(\hat{R}' W \hat{R} \right)^{-1} \hat{R}' W \mathbb{E}_t \mathbf{Y}_0, \quad (1)$$

where $\hat{\delta}^* \in \mathbb{R}^H$ is the vector of optimal policy adjustments across horizons $h = 0, \dots, H-1$, $\mathbb{E}_t \mathbf{Y}_0 \in \mathbb{R}^{2H}$ is the stacked vector of conditional forecasts for inflation and output gap objectives over the H -period horizon, and $\hat{R} \in \mathbb{R}^{2H \times H}$ is the empirical estimate of the impulse response matrix R (in BM, recovered from high-frequency monetary surprises) mapping policy adjustments at each horizon to outcomes.¹

In the baseline linear-quadratic case with no constraints, the OPP has the closed-form solution (1); in constrained environments such as the zero lower bound (Section 3.3), $\hat{\delta}^*$ must be computed via constrained optimization. Crucially, neither the identification of $\hat{R}$ nor the formation of $\mathbb{E}_t \mathbf{Y}_0$ requires knowledge of the full structural model.

This paper asks when that claim of sufficiency holds. Taking BM's algebra as given, we characterize the economic conditions under which the two statistics are genuinely sufficient, and identify three structurally distinct features of macroeconomic environments that violate them—each disabling the *correction* of non-optimality while leaving its *detection* intact. The first concerns the stability of the impulse-response matrix R across policy regimes. BM's correction requires that the R estimated from historical data equals the causal multiplier governing the economy's response under the corrected policy. This equality fails when private-sector expectation formation is endogenous to the policy rule — a manifestation of the Lucas [1976] critique. The relevant structural condition, which BM acknowledge but do not formalize, is *policy-block separability*: the non-policy block of the model must be invariant to the policy rule. When a researcher estimates R from a high-credibility sample and applies it in a low-credibility environment where agents form expectations differently, the two matrices diverge. In our calibrated illustration this divergence is large enough to reverse the sign of the recommended policy adjustment, generating welfare losses that exceed the correctable gain. The second counterexample targets a different limitation: BM's optimality condition $R' W \mathbb{E}_t \mathbf{Y}_0 = 0$ is a gradient, or first-order, condition. It correctly characterizes a local optimum, but in economies admitting multiple rational expectations equilibria the economy may be trapped at an inferior equilibrium where $R' W \mathbb{E}_t \mathbf{Y}_0 = 0$ holds exactly — because agents are optimizing *within* that equilibrium — while a non-marginal policy inter-

¹The relationship between the vector δ and the scalar surprise shock ε_t commonly estimated via high-frequency identification deserves clarification. In empirical applications, researchers typically estimate impulse responses to a one-time surprise $\varepsilon_0 = 1$ at the initial period, with all future policy determined by the systematic rule. This corresponds to $\delta = (1, 0, \dots, 0)'$ in our notation. More generally, δ allows for horizon-specific adjustments: $\delta = (\delta_0, \delta_1, \dots, \delta_{H-1})'$ represents a sequence of policy corrections, where δ_h is the adjustment to the policy instrument at horizon h relative to the baseline path. The OPP formula (1) optimally chooses this entire path jointly, rather than restricting attention to a single contemporaneous shock. When the policy rule features no serial correlation (as in the simple Taylor rule $i_t = \phi_\pi \pi_t + \phi_x x_t + \varepsilon_t$), the two interpretations coincide for the impact period: $\delta_0 = \varepsilon_0$.

vention could access a globally superior allocation. BM’s test is blind to this possibility by construction. The third counterexample concerns structural nonlinearity. At the zero lower bound, the correction step requires knowledge of which future horizons the constraint binds, what impulse responses govern transmission in the constrained region, and what shadow price the binding constraint carries — none of which are recoverable from the BM’s two sufficient statistics alone. Structural information re-enters through the specification of the constrained problem, quietly undermining the model-free character of the approach. In all three cases the *detection* step remains valid and robust; the failure is specific to the correction.

The paper closest to ours in spirit is Caravello et al. [2026], who ask a related but distinct question: when does the sufficient-statistics approach of BM and McKay and Wolf [2023] require supplementation with structural models? Their answer turns on the yield curve. Existing high-frequency instruments primarily identify causal effects of transitory interest-rate changes — shocks to the short end of the yield curve — while persistent policy counterfactuals require causal effects further out along the curve, which empirical variation cannot deliver without structural extrapolation. Among model features, they show that behavioral frictions in price-setter expectations, not household heterogeneity, drive the extrapolation gap. Our analysis is complementary rather than competing. Caravello et al. take the short-end estimate of R as internally valid and ask what additional structure fills in the rest. We show that even the short-end R is not regime-invariant: it depends on whether private agents believe the central bank’s long-run commitment to the policy path, and this credibility dimension operates precisely through the far-ahead rate expectations that high-frequency identification cannot span. The two papers therefore converge on a common conclusion from different angles: BM’s correction step is incomplete for persistent policy adjustments, because the available empirical evidence is both *incomplete at the long end* [Caravello et al., 2026] and *regime-dependent at the short end* (this paper); and both limitations trace to the same underlying fault line, namely that the commitment–discretion distinction in monetary policy is encoded in the dimensions of the yield curve that surprise-based instruments cannot reach.

The remainder of the paper is organized as follows. Section 2 restates the BM setup to fix notation. Section 3 presents each counterexample in turn, providing both formal conditions and economic intuition. Section 4 quantifies the welfare cost of the correction-step failure through a calibrated simulation of Counterexample A. Section 5 discusses the implications for empirical practice and suggests extensions. Section 6 examines what the detection step itself identifies, distinguishes non-optimality from time inconsistency, and asks whether the OPP, followed as a systematic rule, can serve as a credibility-building commitment device. Section 7 concludes.

2 The Barnichon–Mesters Framework

We restate the BM setup to fix notation and, more importantly, to make explicit the three maintained assumptions on which the framework rests. Each of the counterexamples in Section 3 isolates the failure of exactly one of these assumptions.

2.1 Objects and the two sufficient statistics

A policymaker decides at date t and cares about a vector of objectives—in the monetary application, inflation and the output gap—over an H -period horizon. We stack these objectives into the random vector

$$\mathbf{Y}_0 \equiv (y'_t, y'_{t+1}, \dots, y'_{t+H-1})' \in \mathbb{R}^{2H}, \quad (2)$$

where, following BM, the subscript 0 marks the impact period of the policy horizon as viewed from the decision date t . The policymaker minimizes the expected quadratic loss

$$L_t = \frac{1}{2} \mathbb{E}_t[\mathbf{Y}_0' W \mathbf{Y}_0] = \frac{1}{2} \mathbb{E}_t \|\mathbf{Y}_0\|_W^2, \quad (3)$$

where W is a diagonal matrix and $\|v\|_W^2 \equiv v' W v$ denotes the associated weighted norm. Following our convention throughout, W compresses into a single object both the relative preference weights on the objectives and the discount factors across horizons.²

The policymaker operates within a generic linear environment

$$A_{yy} \mathbb{E}_t \mathbf{Y}_0 - A_{yp} P_t^e = B_{y\xi} \mathbb{E}_t \Xi_t + B_{yx} \mathbf{X}_t^-, \quad (4)$$

where P_t^e is the expected future policy path, Ξ_t are structural shocks, and $\mathbf{X}_t^-$ captures initial conditions. Augmenting the policy instrument by a policy shock ε_t and solving the model delivers the reduced form

$$\mathbf{Y}_0 = R \varepsilon_t + C \xi_t, \quad (5)$$

in which R is the matrix of impulse responses of the objectives to policy shocks and C governs structural-shock propagation.

The two objects on which the entire approach turns are:

- (i) the *conditional forecast* of the objectives given the policy choice, $\mathbb{E}_t \mathbf{Y}_0 \equiv \mathbb{E}_t[\mathbf{Y}_0 \mid \text{policy}]$; and
- (ii) the *impulse responses* R of the objectives to a policy shock, estimated as $\hat{R}$ (in BM, via high-frequency monetary surprises).

²BM keep the variable weights and the discount-factor sequence $\beta = (\beta_0, \beta_1, \dots)'$ as separate objects and absorb $W^{1/2}$ into R and $\mathbf{Y}_0$, writing the optimality condition as $\tilde{R}' \tilde{\mathbf{Y}}_0 = 0$ with tildes denoting pre-weighted objects. We instead carry W explicitly and fold the discount factors into it. The two conventions are mathematically equivalent.

These are BM’s two sufficient statistics. The distinction between the random outcome vector $\mathbf{Y}_0$ and its conditional forecast $\mathbb{E}_t \mathbf{Y}_0$ deserves emphasizing: $\mathbf{Y}_0$ is what the economy realizes, while the sufficient statistic is the forecast $\mathbb{E}_t \mathbf{Y}_0$. Keeping the two distinct—rather than collapsing them into a single symbol—is what makes the two steps below transparent. When we evaluate realized outcomes under a particular regime r in Section 4, we write $\mathbf{Y}^r$.

2.2 Detection and correction

Because $\mathbf{Y}_0$ is linear in the policy shock by (5) and L_t is quadratic, the gradient of the loss with respect to a marginal policy perturbation is $R'W \mathbb{E}_t \mathbf{Y}_0$. The BM equivalence is then

$$i_t = i_t^* \iff R'W \mathbb{E}_t \mathbf{Y}_0 = 0, \quad (6)$$

where i_t^* denotes optimal policy. We call the evaluation of the left-hand object $R'W \mathbb{E}_t \mathbf{Y}_0$ the *detection* step. When it is non-zero, the *correction* step returns the Optimal Policy Perturbation introduced in (1), which in the present weighted notation reads

$$\hat{\delta}^* = - \left(\hat{R}'W\hat{R} \right)^{-1} \hat{R}'W \mathbb{E}_t \mathbf{Y}_0, \quad (7)$$

the adjustment that drives the estimated gradient to zero. BM establish (6) without knowledge of the structural matrices A and B , within a linear-quadratic environment that nests both the Ramsey commitment planner and the Markov-perfect discretionary policymaker as special cases.

2.3 Three maintained assumptions

The validity of the two steps rests on three assumptions, which we state explicitly because each is the target of a counterexample in Section 3.

- (A1) *Linearity.* The mapping from policy to objectives is linear, as in (5), so that a single impulse-response matrix R describes transmission everywhere. *Counterexample C* (§3.3) breaks (A1): at the zero lower bound transmission is state-dependent and no single R is global.
- (A2) *Convexity and a unique optimum.* The loss is convex and the policy problem has a unique optimum, so that the first-order condition (6) is sufficient—not merely necessary—for a *global* optimum. *Counterexample B* (§3.2) breaks (A2): with multiple equilibria the gradient can vanish at a welfare-inferior point.
- (A3) *Policy-block separability (regime-invariance of R).* The non-policy block of the model—and hence R —is invariant to the policy rule, so that the $\hat{R}$ estimated from historical

data equals the R that governs the economy under the corrected policy. *Counterexample A* (§3.1) breaks (A3): when private expectations are endogenous to the regime, $R^C \neq R^D$, a manifestation of the Lucas [1976] critique.

One observation organizes the entire paper. The detection step requires only that $\hat{R}$ identify the local, within-regime gradient; it therefore survives the failure of (A3). The correction step requires in addition that $\hat{R}$ equal the *global, regime-invariant* causal multiplier that will govern the economy once the policy is changed; it therefore fails whenever any of (A1)–(A3) fails. The claimed regime-invariance of the two sufficient statistics is thus not a primitive of the method but an implication of (A1)–(A3): when they hold, BM’s results go through unchanged, and our contribution is to characterize what the correction step does when they do not.

3 Three Counterexamples

3.1 Regime-Dependent Expectation Formation and the Lucas Critique

The most fundamental challenge to the BM framework concerns the stability of the impulse-response matrix R across policy regimes. BM’s identification strategy relies on estimating $\hat{R}$ from historical data—using high-frequency monetary surprises as instruments in their empirical application. The implicit assumption is that the historically estimated $\hat{R}$ is the same R that governs the economy’s response under the corrected policy. This assumption fails when agents’ expectation formation is endogenous to the policy regime.

3.1.1 Setup

Consider the baseline New Keynesian model of BM’s Section II, but allow for two distinct regimes that differ in credibility. Under *Regime C* (commitment), the central bank credibly commits to a future policy path and private agents incorporate this commitment into their inflation expectations via rational forward-looking behavior. The New Keynesian Phillips curve is

$$\pi_t = \beta \mathbb{E}_t^C[\pi_{t+1}] + \kappa x_t + \xi_t, \quad (8)$$

where $\mathbb{E}_t^C$ denotes expectations formed under Regime C.

Under *Regime D* (discretion), the central bank re-optimizes each period and rational agents anticipate this

$$\pi_t = \beta \mathbb{E}_t^D[\pi_{t+1}] + \kappa x_t + \xi_t, \quad (9)$$

where $\mathbb{E}_t^D$ denotes expectations under Regime D. Although equations (8) and (9) appear identical in form, the equilibrium laws of motion for inflation differ because $\mathbb{E}_t^C[\pi_{t+1}] \neq$

$\mathbb{E}_t^D[\pi_{t+1}]$ in general: under commitment, agents believe that the current surprise signals the beginning of a new, persistently tighter policy path, whereas under discretion no such belief is sustained.

3.1.2 Divergence of impulse responses

In models with forward-looking expectations, the propagation of a given surprise shock differs across commitment and discretion regimes due to differences in continuation policy. Under commitment, agents' rational expectations embed the central bank's promised future path; under discretion, they incorporate only period-by-period re-optimization. This generates regime-dependent impulse responses. Let $|R^r|_\pi \equiv \left| \sum_{h=0}^{H-1} \beta^h R_{\pi,h}^r \right|$ denote the magnitude of the cumulative discounted inflation response under regime $r \in \{C, D\}$, respectively. In Appendix A we show analytically that $R^C \neq R^D$ and derive conditions under which cumulative commitment response can exceed the discretion response in absolute value. The magnitude and direction of this difference depend on model structure: Appendix A shows that in the standard NK model the commitment cumulative response exceeds the discretion response ($|R^C|_\pi > |R^D|_\pi$), while Section 4 demonstrates an empirically important hybrid-NKPC case where the relation reverses ($|R^D|_\pi > |R^C|_\pi$ for the inflation channel), leading to a sign reversal in the recommended policy correction.

We emphasize that the regime-dependence mechanism outlined above concerns *systematic* differences in transmission across credibility environments, not the information content of individual surprise shocks. High-frequency monetary surprises isolate innovations orthogonal to current macroeconomic news, which is appropriate for within-regime identification. Our claim is that the same surprise shock generates different impulse responses when estimated on samples spanning distinct credibility regimes—because the propagation mechanism (governed by γ_f in our quantitative illustration below) differs across regimes, not because agents extract regime information from the shock itself.

3.1.3 Empirical Evidence for Regime-Dependent Monetary Transmission

The theoretical mechanism outlined above—that impulse responses R vary endogenously with the policy regime—is not merely a logical possibility but is supported by substantial empirical evidence from the monetary policy literature. We briefly review three strands of this evidence that directly motivate our quantitative exercise in Section 4.

Structural breaks in policy rules and inflation dynamics. Clarida et al. [2000] estimate Taylor-type monetary policy rules for the Federal Reserve across distinct sub-samples and document a sharp structural break in the systematic component of policy coinciding with Paul Volcker's appointment as Fed Chair in 1979. Pre-Volcker (1960–1979), the estimated interest rate response to inflation was $\phi_\pi = 0.83$, statistically indistinguishable from

unity and therefore consistent with passive monetary policy that accommodates inflation shocks. Post-Volcker (1982–1997), the estimated response coefficient jumps to $\phi_\pi = 2.15$, significantly exceeding the Taylor principle threshold and consistent with active disinflation.³

Crucially, Clarida et al. [2000] show that this policy-rule shift coincides with a dramatic change in inflation persistence. In the pre-Volcker sample, inflation exhibits near-unit-root behavior with an autoregressive coefficient of $\rho_\pi \approx 0.95$; in the post-Volcker sample, persistence falls to $\rho_\pi \approx 0.65$. This co-movement of policy activism and reduced-form dynamics is precisely the regime-dependence predicted by forward-looking models: when the central bank credibly commits to aggressive disinflation (the “Volcker shock”), private-sector expectations adjust and inflation becomes less persistent even conditional on the same structural parameters κ , β , σ governing the Phillips curve and IS equation.

Time-varying coefficients in inflation forecasting equations. Cogley and Sargent [2005] estimate a time-varying parameter (TVP) VAR on U.S. inflation, output, and interest rates over 1959–2001 and document systematic drift in the reduced-form coefficients. The impulse response of inflation to a monetary policy shock is not constant: it is large in the 1970s (peak response of 0.8 pp to a 100 bp shock), declines sharply in the early 1980s, and stabilizes at a smaller magnitude (0.4 pp) through the 1990s. The authors attribute this variation to shifts in private-sector beliefs about the Fed’s commitment to price stability—exactly the expectation-formation channel emphasized in our Counterexample A.

Importantly, Cogley and Sargent [2005] show that the time-variation cannot be fully explained by changes in shock volatility or identified structural breaks in the exogenous driving processes. The reduced-form propagation mechanism itself is regime-dependent, which is inconsistent with the assumption that a single impulse-response matrix R estimated over the full sample provides valid identifying information for policy correction in any sub-period.

Direct estimation of expectation-formation parameters. Ireland [2007] structurally estimates a New Keynesian DSGE model with regime-switching in the monetary policy rule and explicitly allows the forward-looking weight γ_f in the hybrid Phillips curve to differ across high- and low-credibility regimes. Using Bayesian methods and data from 1960–2006, he finds that γ_f is significantly higher in periods when the Taylor rule coefficient ϕ_π exceeds 1.5 (the “active policy” regime) than in periods with $\phi_\pi < 1.2$ (the “passive policy” regime). The posterior mean estimates are $\gamma_f^{\text{active}} = 0.71$ versus $\gamma_f^{\text{passive}} = 0.43$, with the difference statistically significant at conventional levels. This finding directly validates the calibration strategy we adopt in Section 4, where we use $\gamma_f^C = 0.88$ for the high-credibility regime and $\gamma_f^D = 0.42$ for the low-credibility regime. The Ireland [2007] estimates confirm that the

³The gap years 1979–1982 are excluded due to the Fed’s non-borrowed reserves operating procedure, which temporarily suspended interest rate targeting.

forward-looking component of expectation formation is indeed endogenous to the perceived policy regime, not an invariant structural parameter.

A complementary dimension of the same problem is documented by Caravello et al. [2026], who show that the empirical instruments used to estimate $\hat{R}$ —including the high-frequency monetary surprises employed by BM—primarily identify causal effects of *transitory* policy rate changes, corresponding to the short end of the yield curve. For persistent or commitment-type policy changes, causal effects at the long end of the yield curve are required but not identified from existing data variation. This empirical limitation maps directly onto the commitment-discretion distinction central to our Counterexample A: under commitment, the Ramsey optimal policy operates through long-dated expected rate changes (the “lower for longer” promise), precisely the dimension that high-frequency identification cannot reach. Under discretion, only short-end effects are operative, and these are what BM estimate. Caravello et al. [2026] further find that the strength of behavioral frictions among price-setters—which parameterize the degree to which agents respond to long-horizon policy commitments—is the key structural feature governing long-end causal effects, a finding that echoes the centrality of the forward-looking weight γ_f in our quantitative illustration.

Summary and implications Together, these three empirical findings establish that monetary transmission—as captured by the impulse-response matrix R —is regime-dependent. The co-movement of policy-rule parameters, inflation persistence, time-varying VAR coefficients, and estimated expectation-formation weights all point to the same conclusion: $\hat{R}^C \neq \hat{R}^D$ in the data, not merely in theory.

This evidence has direct implications for the BM sufficient statistics approach. If a researcher estimates $\hat{R}$ using a sample drawn predominantly from a high-credibility regime (e.g., the Volcker–Greenspan era) and applies the correction formula (1) in a low-credibility environment, the historically estimated impulse responses will systematically misrepresent the true transmission mechanism. Section 4 quantifies the welfare cost of this misspecification through a calibrated simulation, showing that the correction-step failure is not a second-order concern but can reverse the sign of the recommended policy adjustment.

3.1.4 Failure of the correction step

Suppose the econometrician estimates R using a sample drawn predominantly from a credible-commitment regime—the Volcker–Greenspan era in the United States being the natural empirical analogue—obtaining $\hat{R}^C$. The current central bank, however, operates under discretion, perhaps because its credibility has eroded or because it is in an emerging-market context. The BM framework computes the OPP as

$$\delta^{*(C)} = - \left(\hat{R}^C' W \hat{R}^C \right)^{-1} \hat{R}^C' W \mathbb{E}_t \mathbf{Y}_0. \quad (10)$$

When the central bank implements $\delta^{*(C)}$, however, private agents respond according to $R^D \neq R^C$, and the actual change in outcomes is

$$\Delta \mathbb{E}_t \mathbf{Y}_0 = R^D \delta^{*(C)} \neq R^C \delta^{*(C)}. \quad (11)$$

The correction fails to close the welfare gap. In the standard forward-looking New Keynesian model (Appendix A), commitment generates a larger cumulative disinflation response per unit shock ($|R^C|_\pi > |R^D|_\pi$), so the wrong OPP is too conservative. In the empirically motivated hybrid model of Section 4, the direction reverses—the backward-looking regime generates larger persistent inflation responses ($|R^D|_\pi > |R^C|_\pi$ for the inflation channel)—causing the wrong OPP to recommend the incorrect *sign* of adjustment. In both cases, iterating on $\delta^{*(C)}$ with the wrong multiplier may fail to converge or converge to a sub-optimal steady state. The direction and magnitude of the correction-step failure are thus model-dependent; Section 4 quantifies the empirically plausible case.

It should be emphasized that the *detection* step of BM remains valid throughout this example: $\hat{R}^C' W \mathbb{E}_t \mathbf{Y}_0 \neq 0$ correctly signals non-optimality. The concern is specific to the *correction* step, which is precisely the more ambitious part of BM's contribution.

The scenario described above is not merely hypothetical. The practice of estimating monetary policy transmission using samples that span multiple distinct policy regimes is endemic to the empirical macro literature. BM's own empirical application draws on data running from 1990 to 2017, a period that encompasses the high-credibility Greenspan era, the unconventional policy post-2008, and the forward guidance experiments of 2011–2015. Cogley and Sargent [2005] document that the impulse response of inflation to a monetary policy surprise shifts markedly across these sub-periods, consistent with our Counterexample A. If these regimes differ in their R , the identifying assumption underlying the OPP correction is violated.

Proposition 1 (Policy-Block Separability as Necessary Condition for Regime-Invariance). *Let the economy satisfy the linear environment (4). Define the policy-block separability condition as the requirement that A_{yy} and $B_{y\xi}$ are invariant to the parameters of the policy rule.*

- (i) *Sufficiency. Under policy-block separability, BM's equivalence (6) holds for any policy regime, and the OPP (1) is welfare-improving for any initial condition $\mathbb{E}_t \mathbf{Y}_0 \neq 0$.*
- (ii) *Necessity. If policy-block separability fails—so that $A_{yy}(r)$ or $B_{y\xi}(r)$ differ across regimes $r \in \{C, D\}$ —then the impulse-response matrices R^C and R^D differ, and there exist initial conditions $\mathbb{E}_t \mathbf{Y}_0$ and regime transitions from r to r' such that applying the OPP computed under regime r in a regime- r' economy yields strictly higher welfare loss than no correction at all.*

Proof. Part (i). Solving the linear system (4) for the objective vector expresses the impulse-response matrix as a function $R = R(A_{yy}, A_{yp}, B_{y\xi}, B_{yx})$ of the structural blocks, with the policy rule entering only through A_{yp} . Under policy-block separability A_{yy} and $B_{y\xi}$ do not depend on the policy rule, so the R that governs transmission is the same object in the estimation sample and under the corrected policy. The detection identity (6) therefore holds with the operative R , and the loss along the perturbation, $L(\delta) = \frac{1}{2}(\mathbb{E}_t \mathbf{Y}_0 + R\delta)'W(\mathbb{E}_t \mathbf{Y}_0 + R\delta)$, is strictly convex because $R'WR \succ 0$. Its unique minimizer is the OPP (1), which satisfies $L(\hat{\delta}^*) \leq L^0$ with strict inequality whenever $R'W \mathbb{E}_t \mathbf{Y}_0 \neq 0$ —that is, the correction is welfare-improving at every forecast at which the economy is not already at its optimum.

Part (ii). The solution map $R(\cdot)$ is non-constant in the non-policy block, $\partial R / \partial A_{yy} \neq 0$. Hence if separability fails— $A_{yy}(C) \neq A_{yy}(D)$ or $B_{y\xi}(C) \neq B_{y\xi}(D)$ —then, outside a measure-zero set of structural parameters, the induced responses R^C and R^D are not collinear in the inner product $\langle a, b \rangle_W \equiv a'Wb$. Linear independence of R^C and R^D makes the functionals $\mathbb{E}_t \mathbf{Y}_0 \mapsto R^{C'}W \mathbb{E}_t \mathbf{Y}_0$ and $\mathbb{E}_t \mathbf{Y}_0 \mapsto R^{D'}W \mathbb{E}_t \mathbf{Y}_0$ independent, so there exists a forecast $\mathbb{E}_t \mathbf{Y}_0$ with

$$R^{C'}W \mathbb{E}_t \mathbf{Y}_0 > 0 > R^{D'}W \mathbb{E}_t \mathbf{Y}_0.$$

A central bank that estimates R^C on the regime-C sample then applies $\delta^{*(C)} = -R^{C'}W \mathbb{E}_t \mathbf{Y}_0 / \|R^C\|_W^2 < 0$, while the economy transmits through R^D , so the realized loss is

$$L = L^0 + \delta^{*(C)} R^{D'}W \mathbb{E}_t \mathbf{Y}_0 + \frac{1}{2}(\delta^{*(C)})^2 \|R^D\|_W^2.$$

Both correction terms are strictly positive—the linear term because $\delta^{*(C)} < 0$ and $R^{D'}W \mathbb{E}_t \mathbf{Y}_0 < 0$, the quadratic term because $\|R^D\|_W^2 > 0$ —so $L > L^0$: applying the OPP computed under regime C in a regime-D economy yields strictly higher loss than no correction. $\square$ $\square$

Part (i) restates BM's main result within the present notation.⁴ Part (ii) is our contribution: it identifies the necessary condition whose failure generates the counterexamples of Section 3. This proposition does not extend to Counterexamples B and C, which fail for different structural reasons: Counterexample B fails because BM's condition is local and the policy problem is non-convex; Counterexample C fails because the constrained problem imports structural knowledge beyond the two statistics. Policy-block separability, in contrast, is the specific condition whose violation drives the regime-mismatch failure of Counterexample A. In the hybrid New Keynesian model of Section 4, separability fails because the Phillips curve weight γ_f —and hence the propagation matrix A_{yy} —is endogenous to the credibility of the policy rule, as confirmed empirically by Ireland [2007].

⁴BM acknowledge in their web appendix that the non-policy block coefficients A_{yy} and $B_{y\xi}$ must be invariant to the parameters of the policy rule, noting that when it fails—as when inflation expectations depend on the policy regime—the OPP can detect non-optimality but can no longer guarantee improvement. Our counterexamples make this restriction concrete and quantify its empirical bite.

3.2 Multiple Rational Expectations Equilibria

The BM sufficiency result relies on the loss function (3) being convex in the policy instrument and the model (4) being linear. Together these properties ensure that $R'W \mathbb{E}_t \mathbf{Y}_0 = 0$ is not only necessary but also sufficient for a global optimum. Our second counterexample demonstrates that this guarantee evaporates in economies with multiple rational expectations equilibria.

3.2.1 Setup

Consider a stylized model of self-fulfilling inflation expectations, in the spirit of Cagan [1956] and the sunspot literature surveyed by Farmer [1999]. There exist two stationary rational expectations equilibria: a low-inflation equilibrium $\mathcal{E}^*$ and a high-inflation equilibrium $\mathcal{E}^{**}$. Both equilibria are consistent with the same structural parameters; they differ in agents' co-ordination on distinct self-fulfilling beliefs. The loss function is strictly lower at $\mathcal{E}^*$ than at $\mathcal{E}^{**}$: $L(\mathcal{E}^*) < L(\mathcal{E}^{**})$.

3.2.2 The OPP declares optimality at the inferior equilibrium

Suppose the economy is at $\mathcal{E}^{**}$ and the econometrician estimates R^{**} —the impulse responses in the neighborhood of $\mathcal{E}^{**}$. The BM gradient condition evaluated locally at $\mathcal{E}^{**}$ yields

$$R^{**'}W \mathbb{E}_t \mathbf{Y}^{**} = 0, \quad (12)$$

because $\mathcal{E}^{**}$ is, by construction, a rational expectations equilibrium: the central bank is minimizing its loss conditional on agents' equilibrium beliefs, and the first-order conditions are satisfied within that equilibrium. The OPP therefore returns $\delta^* = 0$ and classifies the policy as optimal. Yet the economy is trapped in the welfare-inferior equilibrium, and a large, discrete shift in the policy stance—one sufficient to move agents' co-ordination to $\mathcal{E}^*$ —would generate a significant welfare improvement.

3.2.3 Economic intuition

Impulse responses measure the response of the economy to a small perturbation from the current equilibrium, holding fixed agents' equilibrium selection. They cannot, by construction, detect the availability of a superior equilibrium that requires a non-marginal policy change to access. The BM framework is an inherently local, gradient-based method: its power is limited in environments where the global policy problem is non-convex.

This limitation is of direct empirical relevance. Episodes of high and entrenched inflation—such as the US stagflation of the 1970s and persistent high-inflation regimes documented by Sargent [1982] for several post-war economies—can be understood as economies trapped in

inferior expectations equilibria [Loyo, 1999]. The standard prescription of a large, credible disinflation is precisely the kind of non-marginal intervention that the OPP would fail to recommend, because condition (12) is satisfied at the inferior equilibrium.

In the language of BM, the two sufficient statistics are *sufficient within* an equilibrium but *not across* equilibria.

3.3 The Zero Lower Bound and Structural Nonlinearity

Our third counterexample concerns the correction step in nonlinear environments, with the zero lower bound (ZLB) as the canonical illustration. At the ZLB, forward guidance is precisely the policy instrument of choice—corresponding to non-zero δ_h at future horizons in the notation of footnote 1, rather than a contemporaneous surprise $\varepsilon_0 = 1$. This distinction matters because the high-frequency instruments used to estimate $\hat{R}$ identify the contemporaneous surprise channel, not the forward-guidance channel operative at the ZLB. BM provide a constrained OPP for the ZLB in their web appendix (Example S3, Corollary S4), and we argue that this extension, while internally consistent, quietly re-imports structural knowledge that the authors elsewhere claim is unnecessary.

3.3.1 State-dependence of the impulse-response matrix

The standard impulse response $\hat{R}$ is typically estimated from a sample dominated by conventional-policy periods when the nominal rate was strictly positive. At the ZLB, the causal effect of a policy shock on outcomes is governed by a materially different propagation mechanism. Three channels are affected:

1. **Interest rate channel.** The nominal rate cannot fall below zero, so conventional IS transmission is mechanically attenuated. A commitment to keep rates at zero for longer has a weaker immediate effect on long-term rates the more credible it is that rates will remain low in any case.
2. **Signaling channel.** A surprise easing delivered via forward guidance rather than a contemporaneous rate cut carries different information content than a conventional rate surprise, complicating the interpretation of high-frequency instruments.
3. **Balance-sheet channel.** In heterogeneous-agent New Keynesian (HANK) models [Auclert et al., 2021, McKay and Wolf, 2023], the aggregate policy multiplier depends on the cross-sectional distribution of liquid wealth, which is itself endogenous to the history of policy and may be far from its steady-state value after a prolonged ZLB episode.

Denoting the true ZLB-period impulse response by R^{ZLB} , we have $R^{\text{ZLB}} \neq \hat{R}$ in each of these channels. The constrained OPP in BM’s web appendix handles this by explicitly imposing the ZLB constraint on the policy path:

$$\delta^{*,\text{ZLB}} = \arg \min_{\delta \in \mathbb{R}^H} \frac{1}{2} (\mathbb{E}_t \mathbf{Y}_0 + R\delta)' W (\mathbb{E}_t \mathbf{Y}_0 + R\delta) \quad \text{s.t.} \quad i_{t+h} + \delta_h \geq 0 \quad \forall h = 0, \dots, H-1, \quad (13)$$

where i_{t+h} denotes the baseline expected policy rate at horizon h (absent any correction), and the constraint requires that the corrected path $\{i_{t+h} + \delta_h\}_{h=0}^{H-1}$ remains non-negative at all horizons. This is a quadratic program with H linear inequality constraints, one for each horizon at which the ZLB may bind.

The construction of the feasible set in (13) requires structural knowledge beyond BM’s two sufficient statistics. While *observing* that the nominal rate is at zero requires no model (point i), three substantive difficulties remain:

- (i) *Forecasting constraint duration:* Determining at which future horizons h the constraint will bind requires a model of the policy path and the economy’s evolution.
- (ii) *State-dependent transmission:* The true impulse response matrix at the ZLB, R^{ZLB} , differs from the conventional-policy estimate $\hat{R}$ due to the channels outlined above. The researcher must know *which* impulse response matrix governs transmission when the constraint binds.
- (iii) *Constraint shadow price:* The Lagrange multiplier on the ZLB constraint in (13) depends on the structural model and cannot be recovered from reduced-form estimates of $\hat{R}$ alone.

Each of these requires information outside BM’s two sufficient statistics, quietly re-importing the structural knowledge the framework sought to avoid.

The same difficulty applies in HANK models. BM and McKay and Wolf [2023] argue that the framework extends to HANK, but the extension holds only when the aggregate impulse response is invariant to the wealth distribution. After a prolonged low-rate environment that concentrates liquid wealth among high-marginal-propensity-to-consume households, an estimated $\hat{R}$ drawn from tranquil periods understates the contractionary power of a rate increase via the income channel. The correction step (1) will again recommend too small an adjustment.

4 A Quantitative Illustration of Counterexample A

The theoretical argument of Section 3.1 raises the question of whether the correction-step failure is a second-order nuisance or a quantitatively important concern. To address this,

we simulate the hybrid New Keynesian model, compute BM’s OPP under each regime, and measure the welfare cost of applying the correction with the wrong impulse-response matrix. We find that the failure is not second-order: under our baseline calibration, applying the misspecified OPP is strictly worse than making no correction at all.

4.1 Simulation Design

4.1.1 Model

We work with the three-equation New Keynesian model of BM’s Section 2, extended to allow the Phillips curve to be hybrid rather than purely forward-looking [Galí and Gertler, 1999]:

$$\pi_t = \gamma_f \beta \mathbb{E}_t[\pi_{t+1}] + \gamma_b \pi_{t-1} + \kappa x_t + \xi_t, \quad (14)$$

$$x_t = \mathbb{E}_t[x_{t+1}] - \frac{1}{\sigma}(i_t - \mathbb{E}_t[\pi_{t+1}]), \quad (15)$$

$$i_t = \phi \pi_t + \varepsilon_t. \quad (16)$$

The parameter $\gamma_f \in (0, 1)$ governs the weight on forward-looking inflation expectations in the Phillips curve, and $\gamma_b = 1 - \gamma_f$ governs the weight on lagged inflation. When $\gamma_f = 1$ the model collapses to the standard purely forward-looking New Keynesian Phillips curve; when $\gamma_f < 1$ inflation exhibits endogenous inertia driven by past realizations.

The forward-looking weight γ_f serves as the identifying dimension for our two regimes. Under a credible commitment regime, price-setters base their decisions predominantly on expected future marginal costs and central bank announcements, generating a high γ_f . Under a low-credibility discretionary regime, price-setters place more weight on backward-looking rules of thumb, generating a lower γ_f . This mapping between credibility and γ_f is grounded in the micro-founded hybrid model of Galí and Gertler [1999], who derive γ_f from the share of forward-looking optimizing firms; a lower share corresponds to an environment in which firms are less responsive to CB communication about future policy.

4.1.2 Regimes and calibration

We consider two regimes:

- **Regime C (commitment / high credibility):** $\gamma_f^C = 0.88$. This value is consistent with the purely forward-looking limit used in Woodford [2003] and the calibrations in Galí [2015]. It represents an environment in which the central bank’s announcements are highly credible and agents assign near-full weight to the bank’s stated future policy path.
- **Regime D (discretion / low credibility):** $\gamma_f^D = 0.42$. This value is taken directly from the empirical estimates of Galí and Gertler [1999], who find $\gamma_f \approx 0.40$ – 0.45 for

the backward-heavy specification of the hybrid NKPC estimated on post-war US data. It represents an environment in which past inflation is nearly as influential as forward expectations—the situation prevailing, for instance, during the pre-Volcker period or in emerging market economies with imperfect CB credibility.

Remaining parameters are set at standard quarterly calibrations: $\beta = 0.99$, $\kappa = 0.15$, $\sigma = 1.0$, $\phi = 1.5$, equal loss weights $\lambda = 1$, and a horizon of $H = 12$ quarters.

4.1.3 Impulse response computation

For each regime we compute the vector of impulse responses $R^r = (\pi_0^r, \dots, \pi_{H-1}^r, x_0^r, \dots, x_{H-1}^r)'$ to a unit policy shock $\varepsilon_0 = 1$ by iterated projection. Because the hybrid NKPC (14) is a second-order difference equation in π_t , the system cannot be solved by simple forward substitution; instead, at each step t we solve simultaneously for (π_t, x_t) conditional on current guesses for all other periods, and iterate to a fixed point. Convergence is achieved to a tolerance of 10^{-14} within 20,000 iterations for both regimes. The solution for R^C corresponds to the standard purely forward-looking model in the limit $\gamma_b \rightarrow 0$.

4.1.4 Policy scenario and the OPP

We consider a tightening episode in which inflation currently stands 2 pp above target and the output gap is -0.5 pp. Baseline conditional forecasts in the absence of any policy correction decay geometrically: $\pi_{0+h} = 2.0 \times 0.85^h$ and $x_{0+h} = -0.5 \times 0.75^h$, yielding the stacked vector $\mathbb{E}_t \mathbf{Y}_0 \in \mathbb{R}^{2H}$.

We compute two versions of BM's OPP (1), using the welfare-weighted form appropriate for a loss function with $\lambda = 1$ and discount factor β :

$$\delta^{*(r)} = -\frac{R^{r'} W \mathbb{E}_t \mathbf{Y}_0}{R^{r'} W R^r}, \quad r \in \{C, D\}, \quad (17)$$

where $W = \text{diag}(\beta^0, \dots, \beta^{H-1}, \lambda\beta^0, \dots, \lambda\beta^{H-1})$ is the diagonal weight matrix.

The *correct* OPP is $\delta^{*(D)}$, computed using the true regime-D impulse responses. The *wrong* OPP is $\delta^{*(C)}$, computed using regime-C impulse responses as the identifying objects—replicating the scenario in which a central bank operating in a low-credibility environment uses an R estimated from a high-credibility historical sample.

Regardless of which OPP was computed, the realized response of the economy always follows Regime D, since that is the true structural environment. The post-correction outcome vector is therefore $\mathbf{Y}^r = \mathbb{E}_t \mathbf{Y}_0 + \delta^{*(r)} R^D$ for each $r \in \{C, D\}$, and welfare is evaluated as $L^r = \frac{1}{2} \mathbf{Y}^{r'} W \mathbf{Y}^r$.

4.2 Results

Table 1 reports the key quantitative results. Figure 1 plots the impulse response functions and the welfare dynamics; Figure 2 shows robustness across the full range of possible γ_f^D values.

4.2.1 Why the two impulse-response matrices differ

Panel A of Table 1 reveals that R^C and R^D have strikingly different shapes, not merely different magnitudes. Under Regime C, the inflation response to a unit rate hike is concentrated at impact (-0.131 pp at $t = 0$) and then decays to near zero within two quarters. The cumulative discounted inflation response is -0.147 . Under Regime D, the impact response is similar (-0.138 pp), but the backward-looking term in (14) generates persistent propagation: the response remains economically significant through $t + 4$ and the cumulative discounted total is -0.291 , nearly twice as large as under Regime C.

The output-gap responses show the reverse pattern. Regime C produces a large impact contraction in output (-0.795 pp) that dissipates almost immediately. Regime D generates a smaller impact (-0.715 pp) but, more importantly, the endogenous inflation persistence under (14) means output rebounds to small positive values at $t + 1$ and beyond. The cumulative discounted output response under Regime C is -0.772 , more than twice the Regime-D value of -0.397 .

The key takeaway is that the two matrices differ not just in scale but in their relative weighting of the inflation and output-gap objectives. Under Regime C, a rate hike is relatively costly to output and modestly beneficial to inflation. Under Regime D, a rate hike is less costly to output (smaller cumulative x response) and more beneficial to inflation (larger cumulative π response). This reversal in the balance of costs and benefits is what drives the dramatic result in Panel B.

4.2.2 The OPP recommends opposite policies under the two regimes

Panel B of Table 1 shows the most striking finding: the two OPPs recommend policies of opposite sign. The correct OPP, $\delta^{*(D)} = +0.413$ pp (+41 basis points), recommends a rate hike. The wrong OPP, $\delta^{*(C)} = -0.155$ pp (−16 basis points), recommends a rate cut.

The economic logic is transparent. The OPP formula (17) is driven by the sign and magnitude of $R'W\mathbb{E}_t\mathbf{Y}_0$, the covariance between the impulse responses and the baseline forecast. Under Regime D, the large cumulative inflation response means that the benefit of tightening (reducing elevated inflation) dominates the cost (widening a modestly negative output gap). The gradient points toward tightening. Under Regime C, the cost–benefit calculus is reversed: the large output-gap response makes further output contraction very costly, while the modest inflation response makes the disinflation benefit small. The gradient mistakenly points toward easing.

In other words, the misspecification is not simply one of magnitude—it is one of *direction*. Using R^C when the true regime is D leads the BM framework to prescribe the wrong sign of the policy adjustment, not merely a scaled-down version of the right one.

4.2.3 Welfare consequences

Proposition 2 (Welfare-Reversing Correction-Step Failure). *Let $\mathbb{E}_t \mathbf{Y}_0 \in \mathbb{R}^{2H}$ be the vector of conditional forecasts under no policy correction, and suppose the true economy operates in Regime D with impulse-response matrix R^D . The correct OPP is $\delta^{*(D)}$ from (17), and the wrong OPP is $\delta^{*(C)}$ computed using $R^C \neq R^D$. Define the realized outcome under regime r as $\mathbf{Y}^r = \mathbb{E}_t \mathbf{Y}_0 + \delta^{*(r)} R^D$ and the welfare loss as $L^r = \frac{1}{2} \mathbf{Y}^{r'} W \mathbf{Y}^r$.*

- (i) *Sign reversal. If $\text{sign}(R^{C'} W \mathbb{E}_t \mathbf{Y}_0) \neq \text{sign}(R^{D'} W \mathbb{E}_t \mathbf{Y}_0)$, then $\delta^{*(C)}$ and $\delta^{*(D)}$ have opposite signs.*
- (ii) *Welfare reversal. Under the condition in (i), $L^C > L^0 > L^D$: the wrong OPP strictly increases the welfare loss relative to no correction.*
- (iii) *Unbounded shortfall. The welfare cost of misspecification, $L^C - L^D$, relative to the correctable gain $L^0 - L^D$, depends only on the ratio of the two corrections:*

$$\frac{L^C - L^D}{L^0 - L^D} = \left(1 - \frac{\delta^{*(C)}}{\delta^{*(D)}}\right)^2.$$

This ratio exceeds 1 whenever $\delta^{(C)}$ and $\delta^{*(D)}$ have opposite signs (the case in part (i)), and grows without bound as $|\delta^{*(C)}/\delta^{*(D)}| \rightarrow \infty$ —in particular when the misspecified response is weak, $\|R^C\|_W \rightarrow 0$, so that the wrong OPP prescribes an arbitrarily large correction.*

Proof. Part (i). The OPP under regime r satisfies $\delta^{*(r)} = -R^{r'} W \mathbb{E}_t \mathbf{Y}_0 / R^{r'} W R^r$. The denominator $R^{r'} W R^r = \|R^r\|_W^2 > 0$ for any $R^r \neq 0$. Hence the sign of $\delta^{*(r)}$ equals minus the sign of $R^{r'} W \mathbb{E}_t \mathbf{Y}_0$. If the gradients $R^{C'} W \mathbb{E}_t \mathbf{Y}_0$ and $R^{D'} W \mathbb{E}_t \mathbf{Y}_0$ have opposite signs, then $\delta^{*(C)}$ and $\delta^{*(D)}$ have opposite signs. $\square$

Part (ii). The welfare loss under the wrong OPP is

$$\begin{aligned} L^C &= \frac{1}{2} (\mathbb{E}_t \mathbf{Y}_0 + \delta^{*(C)} R^D)' W (\mathbb{E}_t \mathbf{Y}_0 + \delta^{*(C)} R^D) \\ &= L^0 + \delta^{*(C)} R^{D'} W \mathbb{E}_t \mathbf{Y}_0 + \frac{1}{2} (\delta^{*(C)})^2 \|R^D\|_W^2. \end{aligned}$$

Since $\delta^{*(C)}$ and $\delta^{*(D)}$ have opposite signs, $\delta^{*(C)}$ has the same sign as $R^{D'} W \mathbb{E}_t \mathbf{Y}_0$, so the linear term $\delta^{*(C)} R^{D'} W \mathbb{E}_t \mathbf{Y}_0 > 0$. It follows that $L^C > L^0$. The inequality $L^0 > L^D$ holds because $\delta^{*(D)}$ is the unique unconstrained minimizer of $L(\delta)$. $\square$

Part (iii). Write $\delta_r \equiv \delta^{*(r)}$ and use $R^D W \mathbb{E}_t \mathbf{Y}_0 = -\delta_D \|R^D\|_W^2$, which follows from the definition of δ_D in (17). Substituting into the quadratic loss gives, for any scalar δ ,

$$\frac{1}{2}(\mathbb{E}_t \mathbf{Y}_0 + \delta R^D)' W (\mathbb{E}_t \mathbf{Y}_0 + \delta R^D) = L^0 + \frac{1}{2} \|R^D\|_W^2 \delta (\delta - 2\delta_D).$$

Evaluating at $\delta = \delta_C$ and $\delta = \delta_D$ and differencing,

$$L^C - L^D = \frac{1}{2} \|R^D\|_W^2 (\delta_C - \delta_D)^2, \quad L^0 - L^D = \frac{1}{2} \|R^D\|_W^2 \delta_D^2,$$

the second being the standard projection gain, strictly positive when $\delta_D \neq 0$. Their ratio is $(1 - \delta_C/\delta_D)^2$. When the two corrections have opposite signs, $\delta_C/\delta_D < 0$ and the ratio exceeds 1; holding the true regime and forecast fixed, $\delta_C = -R^C W \mathbb{E}_t \mathbf{Y}_0 / \|R^C\|_W^2$ diverges as $\|R^C\|_W \rightarrow 0$ with $R^C W \mathbb{E}_t \mathbf{Y}_0$ bounded away from zero, so the ratio grows without bound. $\square$

Remark 1. *The calibrated values in Table 1 instantiate Proposition 2: the gradient sign reversal condition in (i) holds at the baseline calibration, confirming the sign reversal reported in Panel B. The shortfall ratio in (iii) equals 189% in the baseline, consistent with the welfare deterioration reported in Panel C.*

Panel C of Table 1 quantifies the damage. Starting from a welfare loss of 7.18 (discounted quadratic loss) in the no-correction baseline, the correct OPP reduces the loss to 7.13, a gain of 0.049. The wrong OPP, however, increases the loss to 7.22, a deterioration of 0.043 relative to no correction. The total welfare cost of the misspecification is 0.092, amounting to 189% of the correctable welfare gain. Applying the wrong OPP is strictly worse than doing nothing.

The mechanism is intuitive. The wrong OPP cuts rates by 16 bp in an economy with above-target inflation. This easing feeds through the Regime-D impulse responses (which the economy actually obeys), generating a small further increase in inflation at impact. The loss from this inflationary mistake exceeds the loss from the original suboptimal policy, because the correction pushes the economy further from the optimum rather than toward it.

4.2.4 Robustness

Figure 2 plots the welfare shortfall—defined as the welfare loss from using R^C as a fraction of the correctable gain—across the full range of true γ_f^D values from 0.20 to 0.86. By Proposition 2(iii) this shortfall equals $(1 - \delta^{*(C)}/\delta^{*(D)})^2$ and is therefore always non-negative. Several patterns are notable. First, the shortfall exceeds 100% (meaning the wrong OPP is worse than no correction) across essentially the entire range, falling below 100% only as γ_f^D approaches γ_f^C . Second, the shortfall rises monotonically as γ_f^D increases toward $\gamma_f^D \approx 0.73$, where it diverges: at that point the correct OPP is zero—the economy is already at its regime-D optimum given the baseline forecast—so the correctable gain vanishes and any

Table 1: Quantitative Illustration of BM Correction-Step Failure

	Quarter								Cumul.
	$t + 0$	$t + 1$	$t + 2$	$t + 3$	$t + 4$	$t + 6$	$t + 8$	$t + 10$	(disc.)
Panel A: Impulse responses to unit policy shock (pp per pp rate change)									
π Regime C	−0.131	−0.014	−0.001	0.000	0.000	0.000	0.000	0.000	−0.147
π Regime D	−0.138	−0.073	−0.039	−0.021	−0.011	−0.003	−0.001	0.000	−0.291
x Regime C	−0.795	+0.022	+0.002	0.000	0.000	0.000	0.000	0.000	−0.772
x Regime D	−0.715	+0.152	+0.081	+0.043	+0.023	+0.007	+0.002	0.000	−0.397
Ratio $ R^D / R^C $ (π , discounted): 1.98									
Ratio $ R^D / R^C $ (x , discounted): 0.51									
Panel B: OPP recommended policy change (pp; + = rate hike)									
Correct OPP $\delta^{(D)}$ [true regime R^D]: +0.413 pp (+41.3 bp)									
Wrong OPP $\delta^{(C)}$ [misspecified R^C]: −0.155 pp (−15.5 bp) [sign reversal]									
Total shortfall (correct − wrong): 56.8 bp (138% of correct adjustment)									
Panel C: Welfare outcomes (discounted quadratic loss)									
No correction: 7.180									
Correct OPP $\delta^{(D)}$: 7.132 [welfare gain: 0.049]									
Wrong OPP $\delta^{(C)}$: 7.223 [welfare loss vs. baseline: 0.043]									
Welfare cost of misspecification: 0.092 (189% of correctable gain)									

Notes. Standard NK model with hybrid NKPC (14)–(16). Parameters: $\kappa = 0.15$, $\sigma = 1.0$, $\phi = 1.5$, $\lambda = 1.0$, $\beta = 0.99$, $H = 12$ quarters. Regime C: $\gamma_f = 0.88$ [Woodford, 2003, Galí, 2015]. Regime D: $\gamma_f = 0.42$ [Galí and Gertler, 1999], backward-heavy specification). Policy scenario: $\pi_0 = +2.0$ pp, $x_0 = -0.5$ pp, persistence $\rho_\pi = 0.85$, $\rho_x = 0.75$. Realized outcomes always follow Regime-D transmission (R^D) regardless of which OPP was computed. “Cumulative (disc.)” denotes the discounted sum $\sum_{h=0}^{H-1} \beta^h R_{\pi,h}^r$, where $R_{\pi,h}^r$ is the inflation impulse response at horizon h under regime $r \in \{C, D\}$. The welfare loss from the wrong OPP exceeds the correctable gain, implying the misspecified correction is strictly worse than no correction at all.

nonzero correction is infinitely costly in relative terms. Third, the shortfall declines to zero only as $\gamma_f^D \rightarrow \gamma_f^C = 0.88$, confirming that the failure vanishes only when the two regimes become identical. Across the empirically plausible low-credibility range— γ_f^D well below γ_f^C , where the mismatch between the estimated R^C and the true R^D is largest—the wrong OPP is uniformly welfare-reducing.

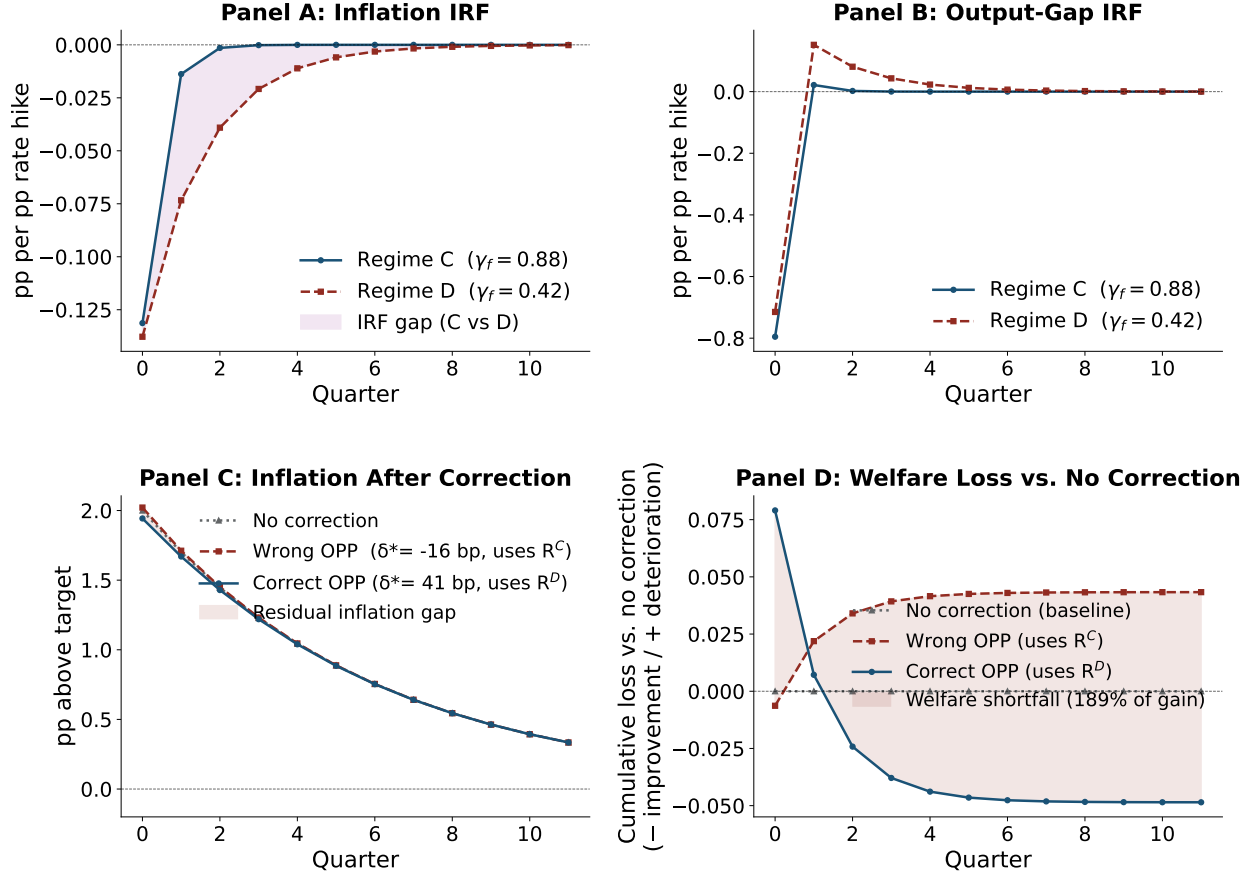


Figure 1: BM Correction-Step Failure Under Regime-Dependent Expectations

Notes. Hybrid New Keynesian model of Section 4: Regime C ($\gamma_f = 0.88$, commitment) versus Regime D ($\gamma_f = 0.42$, discretion); calibration $\kappa = 0.15$, $\sigma = 1$, $\phi = 1.5$, $H = 12$ quarters. **Panel A:** Impulse responses of inflation to a unit policy shock under Regime C ($\gamma_f = 0.88$, solid) and Regime D ($\gamma_f = 0.42$, dashed). Regime D's backward-looking component generates far greater inflation persistence. **Panel B:** Impulse responses of the output gap. Regime C has a larger peak output contraction that dissipates quickly; Regime D has a smaller but more protracted contraction, with modest positive rebounds thereafter. **Panel C:** Inflation paths following the no-correction baseline (dotted), the wrong OPP $\delta^*(C)$ (dashed, red), and the correct OPP $\delta^*(D)$ (solid, blue). The wrong OPP raises inflation above the no-correction path, as the misguided rate cut worsens the inflation gap. **Panel D:** Cumulative discounted welfare loss of each correction relative to the no-correction baseline (the zero line): negative values denote welfare improvement, positive values deterioration. The correct OPP (blue) settles well below zero while the wrong OPP (red) settles above it; the shaded band is the welfare shortfall, reaching 189% of the correctable gain. The correct OPP's transient rise at impact reflects the short-run output cost of tightening, which the subsequent disinflation more than offsets.

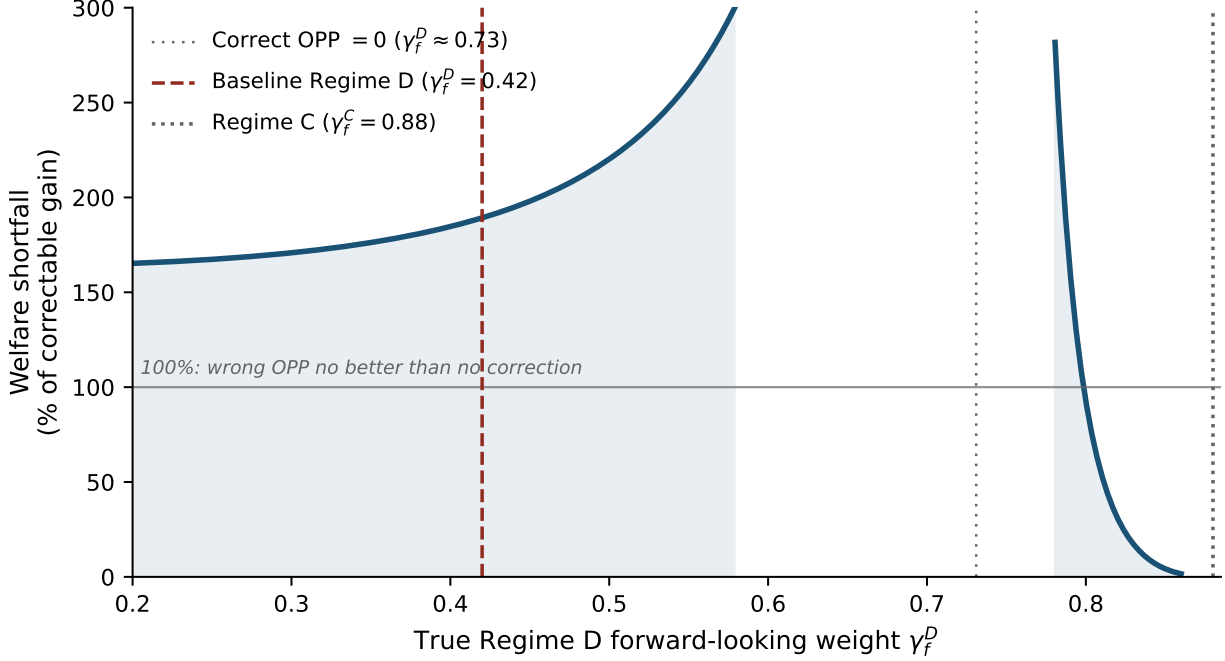


Figure 2: Welfare Shortfall as a Function of the True Forward-Looking Weight γ_f^D

Notes. The curve plots the welfare shortfall from applying the wrong OPP $\delta^{*(C)}$ —computed using Regime-C impulse responses—as a percentage of the correctable welfare gain under the correct OPP $\delta^{*(D)}$, as the true forward-looking weight γ_f^D varies with $\gamma_f^C = 0.88$ held fixed. By Proposition 2(iii) the shortfall equals $(1 - \delta^{*(C)}/\delta^{*(D)})^2$ and is hence non-negative; the vertical axis is capped at 300%. Points above the 100% line (gray) indicate that the wrong OPP is welfare-decreasing relative to no correction. The shortfall diverges at $\gamma_f^D \approx 0.73$ (black dotted line), where the correct OPP is zero—the economy is already optimal under Regime D given the baseline forecast, so the correctable gain vanishes—and the line is suppressed where it leaves the plotting range. The red dashed line marks the Galí-Gertler (1999) calibration ($\gamma_f^D = 0.42$); the shortfall declines to zero only as $\gamma_f^D \rightarrow \gamma_f^C = 0.88$.

5 Discussion: Scope Conditions and Extensions

The three counterexamples share a common structure. Each identifies a condition under which the impulse-response matrix R , as estimable from reduced-form data, fails to equal the structural causal object that the BM correction step requires. In Counterexample A (§3.1), R depends on the policy regime through agents’ beliefs; Section 4 shows this failure is not second-order but can reverse the sign of the recommended policy and generate welfare losses exceeding the correctable gain. In Counterexample B (§3.2), R is estimated at a locally stable but globally inferior equilibrium. In Counterexample C (§3.3), R is estimated away from the nonlinear region where correction is needed.

We emphasize that these observations do not undermine BM’s *detection* result. The condition $R'W \mathbb{E}_t \mathbf{Y}_0 \neq 0$ remains a valid test of local non-optimality across all three counterexamples, provided R is estimated consistently—which it is in cases A and B, though not

necessarily in C. The concern is with the *correction* step, which is the more ambitious part of BM’s contribution. We formalise this asymmetry in Corollary 2.1 below.

Corollary 2.1 (Robustness of the Detection Step). *For any impulse-response matrix R^r that consistently estimates the within-regime response, the condition $R^{r'}W\mathbb{E}_t\mathbf{Y}_0 \neq 0$ is a valid test of local non-optimality: it correctly signals the existence of a welfare-improving marginal policy correction within the prevailing regime, regardless of whether R^r coincides with the cross-regime matrix required for optimal correction. The detection result is therefore robust to the regime misidentification that invalidates Proposition 2.*

Proof. For any R^r consistently estimating the within-regime impulse response and any $\mathbb{E}_t\mathbf{Y}_0 \neq 0$ such that $R^{r'}W\mathbb{E}_t\mathbf{Y}_0 \neq 0$, the correction $\delta^{*(r)} = -R^{r'}W\mathbb{E}_t\mathbf{Y}_0/R^{r'}WR^r$ is a local welfare improvement *within* regime r : applying it reduces L at the margin. This holds regardless of whether regime r is the true structural regime, because it requires only that R^r identifies the local gradient, not the global transmission mechanism. $\square$ $\square$

Extension 1: Regime-robust OPP. The natural analogue of the Hansen and Sargent [2008] robust-control approach would construct a correction that is welfare-improving under the worst-case R in the identified set:

$$\delta^{*,\text{robust}} = \arg \min_{\delta} \max_{R \in \mathcal{R}} \frac{1}{2} (\mathbb{E}_t\mathbf{Y}_0 + R\delta)'W(\mathbb{E}_t\mathbf{Y}_0 + R\delta), \quad (18)$$

where $\mathcal{R}$ is a confidence set for the impulse responses. This conservative correction trades off sharpness for robustness to regime uncertainty, and its solution takes a familiar shrinkage form.

Extension 2: Equilibrium-selection diagnostics. Addressing the multiple-equilibria counterexample requires moving beyond first-order methods. One approach would couple the OPP with a test for equilibrium multiplicity using survey-based inflation expectations data that BM already employ for conditional forecasts. A persistent upward drift in long-run inflation expectations, conditional on the current policy path, would signal the economy approaching an inferior equilibrium and call for a non-marginal policy intervention outside the OPP’s local scope.

Extension 3: State-dependent impulse responses. For the ZLB and HANK environments, the most tractable extension would make R explicitly state-dependent via nonlinear local projections [Jordà, 2005, Ramey and Zubairy, 2018, González-Garzón and Lieb, 2023]. A function $R(s)$, where s indexes the economic regime (ZLB/non-ZLB, low/high household debt), would calibrate the OPP to the local nonlinear environment without committing to a full structural model. Caravello et al. [2026] operationalize a disciplined version of this

strategy that speaks directly to the regime-dependence failure documented in Section 3.1. Their hybrid approach uses empirical causal effects for transitory rate changes—where high-frequency instruments provide valid short-end identification—and supplements with minimal structural extrapolation for the far-dated rate expectations that existing data variation cannot span. Crucially, they find that household heterogeneity (HANK) leaves the long-end extrapolation largely unchanged relative to a representative-agent benchmark, while behavioral frictions in price-setter expectations are decisive [Gabaix, 2020]. This finding maps directly onto the forward-looking weight γ_f in our hybrid NKPC (14): weaker behavioral discounting (high γ_f , *Regime C*) implies that long-end forward guidance is effective and the commitment impulse response R^C is large; stronger discounting (low γ_f , *Regime D*) renders the long end inert and collapses R^C toward R^D , precisely the mechanism driving the welfare reversal in Proposition 2. A productive synthesis of the two approaches would (i) estimate $\hat{R}^{\text{short}}$ from high-frequency instruments as Caravello et al. propose, (ii) use a behaviorally calibrated model to extrapolate to $\hat{R}^{\text{long}}$ with uncertainty bounds reflecting the range of γ_f values documented by Ireland [2007], and (iii) apply the regime-robust OPP of Extension 1 over the resulting uncertainty set $\mathcal{R}$. This three-step strategy would address the two complementary limitations of BM identified in this paper and in Caravello et al. [2026] simultaneously: the identified $\hat{R}$ is both *incomplete at the long end* [Caravello et al., 2026] and *regime-dependent at the short end* (Proposition 1), and both limitations trace to the same fault line—that the commitment–discretion distinction in monetary policy operates through the yield-curve dimensions that surprise-based instruments cannot span.

6 Detection, Time Inconsistency, and the OPP as a Commitment Device

The analysis of Sections 3–4 focused on the correction step of the BM framework. We now examine a deeper question that the correction-step failures expose: what does the detection step actually identify, and can the framework, properly interpreted, speak to time inconsistency as well as to within-regime non-optimality?

6.1 The Detection Step as a Within-Regime Test

The condition $R'W \mathbb{E}_t \mathbf{Y}_0 \neq 0$ identifies something precise and robust: given the policy regime currently prevailing—whatever that regime is—the central bank is not at a local optimum of its own loss function. This is a strictly within-regime statement. It says that, taking as given the equilibrium beliefs of private agents, there exists a marginal policy adjustment that improves welfare.

The result holds regardless of whether agents are purely forward-looking or hybrid, re-

regardless of whether the policymaker operates under commitment or discretion, and regardless of whether the loss function is BM’s quadratic specification or any convex generalization. The detection step is, in this respect, the most durable part of the BM contribution.

However, the within-regime character of the detection step is also its fundamental limitation: it cannot answer the question of whether the prevailing regime is itself optimal. To see why, note that both the Ramsey optimal allocation and the Markov-perfect discretionary equilibrium satisfy $R'W \mathbb{E}_t \mathbf{Y}_0 = 0$ —the former because the Ramsey planner is globally maximizing welfare subject to the implementability constraints, the latter because the discretionary policymaker is locally maximizing welfare taking private-sector beliefs as given.

At the Barro and Gordon [1983] steady state, for instance, $R'W \mathbb{E}_t \mathbf{Y}_0 = 0$ holds exactly: the central bank has no marginal incentive to surprise-inflate, because agents have already priced in the inflation bias. BM’s test would declare this policy optimal.

The welfare loss from the inflation bias is not second-order. Calibrated estimates in the literature [Ireland, 1997, Chari and Kehoe, 1990] put the steady-state consumption-equivalent cost at one to two percent per annum. This loss arises not from any deviation of the gradient condition from zero, but from the equilibrium itself being at the wrong point. Time inconsistency is, in this precise sense, a global pathology that does not register in a first-order test.

6.2 Non-Optimality or Time Inconsistency: Can We Tell the Difference?

The identification problem is sharp. Consider two economies that are observationally equivalent with respect to BM’s two sufficient statistics:

- **Economy 1:** The CB is operating under commitment and is at the Ramsey optimum. R^C and $\mathbb{E}_t \mathbf{Y}_0$ are both consistent with $R^C W \mathbb{E}_t \mathbf{Y}_0 = 0$.
- **Economy 2:** The CB is operating under discretion and is at the Markov-perfect equilibrium. R^D and $\mathbb{E}_t \mathbf{Y}_0$ are both consistent with $R^D W \mathbb{E}_t \mathbf{Y}_0 = 0$.

In both economies the two sufficient statistics satisfy the BM optimality condition. Yet Economy 2 may exhibit a persistent inflation bias, history independence of policy, and welfare losses relative to Economy 1 that are quantitatively large. BM’s framework cannot distinguish these two cases because the identifying variation—surprise policy shocks ε_t —affects the economy symmetrically through R regardless of which regime prevails.

The systematic component of policy, which is where the commitment-versus-discretion distinction lives, is by construction orthogonal to the shock-identified impulse responses.

Formally, the impulse response R identifies the response of $\mathbb{E}_t \mathbf{Y}_0$ to the unexpected component of policy. The welfare difference between commitment and discretion resides in

the *systematic* component—the history-dependence of the Ramsey policy rule, the ‘lower-for-longer’ promise after a cost-push shock, the penalization of past target misses. These features are in the null space of the shock-identification strategy. No reduced-form specification of the BM type can recover them without additional structural content.

6.3 Imperfect Commitment and the Missing Sufficient Statistic

Real central banks operate under imperfect commitment: institutional arrangements, statutory mandates, and reputational capital constrain but do not eliminate the time-inconsistency temptation. Following Erceg and Levin [2003], let $\theta_t \in [0, 1]$ denote the credibility stock at time t , with $\theta_t = 1$ representing full commitment and $\theta_t = 0$ pure discretion. Private agents assign weight θ_t to the committed policy rule and weight $(1 - \theta_t)$ to the discretionary best response, so the effective impulse-response matrix governing current transmission is

$$R^{\theta_t} = \theta_t R^C + (1 - \theta_t) R^D. \quad (19)$$

This linear specification is restrictive and serves primarily to illustrate how credibility variation creates identification challenges for the BM approach. Real-world credibility dynamics likely feature nonlinearities, thresholds, and hysteresis that our linear interpolation abstracts from. The complications we derive below—regarding sample averaging and the missing third sufficient statistic—should be understood as illustrative rather than exhaustive, though the core insight that imperfect commitment creates a wedge between estimated and effective impulse responses holds more broadly.

First, because θ_t is time-varying, the historically estimated $\hat{R}$ is a convolution of R^{θ} over the sample, potentially spanning large credibility swings such as the pre- and post-Volcker eras. The estimated $\hat{R}$ corresponds to the sample-average $\bar{\theta}$, which may bear little relation to the current θ_t precisely when policy guidance is most needed.

Second, the OPP correction δ^* is itself a credibility-relevant action: a CB that follows through on the prescribed adjustment signals type and shifts θ_{t+1} upward, while one that deviates signals the opposite. The welfare function therefore has a second argument, the continuation credibility stock, that is not captured by the within-period loss L_t . The full value function is

$$V_t = L_t(\mathbb{E}_t \mathbf{Y}_0, \delta) + \beta V_{t+1}(\theta_{t+1}(\delta)), \quad (20)$$

where $\theta_{t+1}(\delta)$ is the belief-updating function.

BM’s OPP minimizes L_t alone, treating V_{t+1} as invariant to δ . This is myopically correct but dynamically incomplete: it ignores the shadow price of the credibility constraint, which constitutes a *third sufficient statistic* that the BM framework does not identify. Any empirical proxy for this shadow price—the deviation of long-run inflation expectations from target, a time-varying credibility index, or the slope of the term structure of inflation breakevens—

would in principle supplement R and $\mathbb{E}_t \mathbf{Y}_0$ and bring the framework closer to the full optimization problem in (20).

6.4 The OPP as a Commitment Device: A Reappraisal

The preceding analysis invites a pushback that deserves careful engagement. If BM’s logic is precisely “given the world as it is, is policy optimal, and if not, how should it be corrected”, then perhaps part of what it means to correct policy is, in fact, to build credibility. The OPP is not merely a one-shot adjustment; it is, potentially, the first step in a systematic rule. And rules are commitment devices.

The argument can be made formally. Woodford [2003] and Giannoni and Woodford [2017] show that the optimal commitment policy can always be represented as a targeting rule of the form $f(\pi_t, x_t, \pi_{t-1}, \dots) = 0$, where $f(\cdot)$ is derived from the planner’s first-order conditions. BM’s condition $R'W \mathbb{E}_t \mathbf{Y}_0 = 0$ has exactly this structure. If a central bank publicly commits to setting policy each period so that $R'W \mathbb{E}_t \mathbf{Y}_0 = 0$ holds—and is believed to do so—this announcement is itself an institutional commitment of the forecast-targeting type formalized by Svensson [1999], at least over the dimension of policy that $\hat{R}$ identifies: the causal effects of transitory rate changes at the short end of the yield curve [Caravello et al., 2026].

Private agents who observe a CB consistently applying the BM correction will update θ_t upward, shifting R^θ toward R^C , which in turn alters the recommended correction, which further builds credibility. The within-regime identification problem is not exogenous to the correction; it is potentially resolved by the correction itself.

This is a substantive and, in our view, partially correct reappraisal. It implies that the relevant object is not the one-shot OPP but the *sequential OPP rule*

$$\delta_t^* = -\left(R^{\theta_t'} W R^{\theta_t}\right)^{-1} R^{\theta_t'} W \mathbb{E}_t \mathbf{Y}_0, \quad t = 0, 1, 2, \dots, \quad (21)$$

where the CB re-estimates R^{θ_t} each period as credibility evolves. A CB that follows rule (21) transparently, predictably, and without discretionary deviations is precisely engaging in credibility-building through action rather than announcement. The distinction between “following the OPP” and “having a committed rule” collapses if the OPP *is* the rule.

Three qualifications, however, limit the scope of this reappraisal.

First, convergence is not guaranteed. The sequential rule (21) is a fixed-point iteration in (R^{θ_t}, θ_t) . Whether this iteration converges to the Ramsey optimum depends on the learning dynamics of private agents and the pace of credibility accumulation. Erceg and Levin [2003] show that with imperfectly observable CB preferences, the transition can be protracted and non-monotone; the early stages of a credibility-building episode may require contractionary corrections that reduce welfare in the short run, potentially making the sequential rule

dynamically time-inconsistent at intermediate dates even if it is asymptotically optimal.

Second, the rule must be known to be the rule. Credibility accrues from consistent, predictable behavior that agents can model and verify. BM’s framework, as presented, is a decision-support tool rather than a publicly announced commitment mechanism. For the sequential OPP to function as a credibility device, the CB would need to communicate its targeting criterion—namely, that it will always adjust policy to satisfy $R'W \mathbb{E}_t \mathbf{Y}_0 = 0$ —with sufficient clarity and institutional backing for agents to treat it as binding. This requires exactly the kind of institutional design (inflation-target legislation, central bank charters, transparent loss functions) that the sufficient-statistics approach was intended to avoid specifying. The structural commitment technology re-enters through the back door.

Third, history-dependence is absent. The Ramsey optimum under commitment is not characterized by $R'W \mathbb{E}_t \mathbf{Y}_0 = 0$ at each date independently; it is characterized by a history-dependent targeting rule in which past misses cumulate into current obligations [Woodford, 2003]. A CB that re-applies δ_t^* from scratch each period—using only the current-period forecasts $\mathbb{E}_t \mathbf{Y}_0$ and the current-period R^{θ_t} —satisfies a timeless optimality condition but not the history-dependent Ramsey condition. It achieves something closer to Svensson [1999] flexible inflation targeting than to full Ramsey commitment. This is not a negligible loss: the welfare gain from history-dependence (the ‘make-up strategy’ of committing to overshoot after periods of undershooting) can be quantitatively significant, particularly at the zero lower bound [Eggertsson and Woodford, 2003].

The bottom line is nuanced. BM framework does not stand entirely outside the commitment problem: a CB that systematically follows the OPP rule, communicates this, and is believed is engaging in a form of commitment. The framework is therefore richer than a purely within-period test. But the richness comes at a cost: to realize the credibility dividend, the CB must supply precisely the structural inputs—institutional commitment, transparent objectives, and history-dependence—that the sufficient-statistics approach sought to dispense with. The framework’s model-free character and its capacity to serve as a credibility device are, in this sense, in tension. Resolving that tension is a productive direction for future work, and one to which the sequential OPP in (21) could contribute in two complementary ways: by augmenting the within-period correction with a credibility-spread proxy for the missing third sufficient statistic, and by extending the identified $\hat{R}$ to the long end of the yield curve using the hybrid data-plus-structure strategy of Caravello et al. [2026]. Together, these extensions would move the OPP rule from a partial commitment device—one that disciplines the short-run response but leaves the long-run rate path unverifiable—toward the full forecast-targeting commitment that the Ramsey optimum requires.

7 Conclusion

This paper asked a general question: when can optimal macroeconomic policy be evaluated from a handful of sufficient statistics, and when does that informational parsimony break down? The appeal of the sufficient-statistics program is that it ties welfare-relevant policy questions to a few estimable objects rather than a fully specified structural model; Barnichon and Mesters [2023] bring this logic to monetary policy, showing that two such objects—conditional forecasts of the objectives and impulse responses to policy shocks—suffice to *detect* non-optimal policy, and thereby give rigorous foundations to a practice that central banks already pursue informally. Our contribution is to map the boundary of that result: to characterize the conditions under which the two statistics are genuinely sufficient, and the economically important environments in which they are not.

The counterexamples offered in Sections 3–4 delineate the scope of the correction step. The framework is most reliable when the policy regime is stable, the economy is well-approximated by a convex program with a unique optimum, and transmission is away from structural nonlinearities. When these conditions fail, the OPP can reverse the sign of the recommended policy and generate welfare losses exceeding the correctable gain, as the quantitative illustration of Section 4 demonstrates.

Section 6 raises a deeper concern. The detection step— $R'W \mathbb{E}_t \mathbf{Y}_0 \neq 0$ —is a within-regime test: it identifies deviations from the prevailing equilibrium but cannot distinguish non-optimality from time inconsistency. Both the Ramsey optimum and the inferior Markov-perfect equilibrium satisfy $R'W \mathbb{E}_t \mathbf{Y}_0 = 0$, making the Barro-Gordon inflation bias invisible to the gradient condition. Imperfect commitment complicates matters further: the effective R is a credibility-weighted mixture, and the shadow price of the commitment constraint constitutes a third sufficient statistic that the framework does not identify.

A more optimistic reading, however, is also available. A central bank that publicly commits to always setting policy so that $R'W \mathbb{E}_t \mathbf{Y}_0 = 0$ —the sequential OPP rule—is engaging in exactly the kind of targeting-rule commitment that the optimal policy literature advocates. In this interpretation, BM’s framework is not merely a diagnostic tool but a candidate institutional rule that can build credibility through consistent and transparent application.

The tension between the framework’s model-free character and its capacity to serve as a commitment device points to a productive direction for future work: augmenting the two sufficient statistics with an observable proxy for the credibility stock, and extending the OPP to a dynamic rule that accounts for the effect of current corrections on future transmission.

The sufficient-statistics approach to macroeconomic policy has substantial promise. Realizing that promise fully requires acknowledging both what the two statistics identify—within-regime local optimality—and what they do not: the location of the prevailing equilibrium on the welfare frontier, and the shadow cost of the institutional constraints that keep it there.

A Commitment versus Discretion: Analytic Characterization of Impulse Response Differences

A.1 Model Setup

We characterize the impulse response differences between commitment and discretion in the canonical three-equation New Keynesian model. The structural equations are:

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa x_t + u_t, \quad (22)$$

$$x_t = \mathbb{E}_t[x_{t+1}] - \sigma^{-1}(i_t - \mathbb{E}_t[\pi_{t+1}]), \quad (23)$$

$$u_t = \rho_u u_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim \text{i.i.d.}(0, \sigma_\varepsilon^2), \quad (24)$$

where π_t is inflation, x_t is the output gap, i_t is the nominal interest rate, and u_t is a cost-push shock. The central bank minimizes the expected discounted loss:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t L_t, \quad L_t = \frac{1}{2}(\pi_t^2 + \lambda x_t^2) \quad (25)$$

with $\beta \in (0, 1)$ the discount factor and $\lambda > 0$ the relative weight on output-gap stabilization.

A.2 Discretionary Equilibrium

Under discretion, the central bank re-optimizes each period taking private-sector expectations as given. The problem at time t is:

$$\min_{i_t} \frac{1}{2}(\pi_t^2 + \lambda x_t^2) + \beta \mathbb{E}_t[V_{t+1}] \quad (26)$$

subject to (22) and (23), where V_{t+1} is the continuation value function. Crucially, $\mathbb{E}_t[\pi_{t+1}]$ and $\mathbb{E}_t[x_{t+1}]$ are treated as exogenous to the current policy choice.

A.2.1 Markov-Perfect Equilibrium

We conjecture linear decision rules:

$$\pi_t = \alpha_\pi^D u_t, \quad x_t = \alpha_x^D u_t, \quad i_t = \alpha_i^D u_t. \quad (27)$$

Substituting into (22) and using $\mathbb{E}_t[u_{t+1}] = \rho_u u_t$:

$$\alpha_\pi^D u_t = \beta \rho_u \alpha_\pi^D u_t + \kappa \alpha_x^D u_t + u_t, \quad (28)$$

which yields:

$$\alpha_\pi^D(1 - \beta\rho_u) = 1 + \kappa\alpha_x^D. \quad (29)$$

Similarly, from (23):

$$\alpha_x^D = \rho_u\alpha_x^D - \sigma^{-1}(\alpha_i^D - \rho_u\alpha_\pi^D), \quad (30)$$

which gives:

$$\alpha_x^D(1 - \rho_u) = -\sigma^{-1}(\alpha_i^D - \rho_u\alpha_\pi^D). \quad (31)$$

The first-order condition with respect to i_t (via the constraint multipliers) yields the targeting rule:

$$\pi_t + \lambda\kappa x_t = 0. \quad (32)$$

Applying (32) to the conjectured policies:

$$\alpha_\pi^D + \lambda\kappa\alpha_x^D = 0 \quad \Rightarrow \quad \alpha_x^D = -\frac{\alpha_\pi^D}{\lambda\kappa}. \quad (33)$$

Substituting (33) into (29):

$$\alpha_\pi^D = \frac{1}{1 - \beta\rho_u - \kappa^2/\lambda}, \quad (34)$$

and thus:

$$\alpha_x^D = -\frac{1}{\lambda\kappa(1 - \beta\rho_u - \kappa^2/\lambda)}. \quad (35)$$

The impulse responses under discretion are:

$$R_h^{D,\pi} = \alpha_\pi^D \rho_u^h, \quad R_h^{D,x} = \alpha_x^D \rho_u^h, \quad h = 0, 1, 2, \dots \quad (36)$$

A.3 Commitment Equilibrium

Under commitment, the central bank chooses $\{i_t\}_{t=0}^\infty$ at time $t = 0$ to minimize (25) subject to (22)–(23), taking into account that its future actions influence current expectations.

A.3.1 Timeless-Perspective Solution

Following Woodford [2003], we impose the timeless-perspective initial condition to avoid time-zero exploitation of expectations. The Lagrangian is:

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left\{ \frac{1}{2}(\pi_t^2 + \lambda x_t^2) + \phi_t[\pi_t - \beta\pi_{t+1} - \kappa x_t - u_t] + \psi_t[x_t - x_{t+1} + \sigma^{-1}(i_t - \pi_{t+1})] \right\}. \quad (37)$$

First-order conditions:

$$\pi_t : \quad \pi_t + \phi_t - \beta^{-1}\phi_{t-1} = 0, \quad (38)$$

$$x_t : \quad \lambda x_t - \kappa\phi_t + \psi_t - \beta^{-1}\psi_{t-1} = 0, \quad (39)$$

$$i_t : \quad \sigma^{-1}\psi_t = 0. \quad (40)$$

From (40), $\psi_t = 0$ for all t . Equations (38) and (39) simplify to:

$$\phi_t = \beta\phi_{t+1} - \pi_{t+1}, \quad (41)$$

$$\phi_t = \frac{\lambda}{\kappa}x_t + \beta^{-1}\psi_{t-1}. \quad (42)$$

But $\psi_{t-1} = 0$, so:

$$\phi_t = \frac{\lambda}{\kappa}x_t. \quad (43)$$

Substituting into (41) and using the NKPC:

$$\frac{\lambda}{\kappa}x_t = \beta\frac{\lambda}{\kappa}x_{t+1} - \pi_{t+1} = \beta\frac{\lambda}{\kappa}x_{t+1} - \beta\pi_t + \kappa x_t + u_t. \quad (44)$$

Rearranging:

$$\pi_t = -\frac{\lambda}{\beta\kappa}(x_t - \beta x_{t+1}) - \frac{u_t}{\beta} - \frac{\lambda}{\beta\kappa}x_t. \quad (45)$$

A.3.2 Solution Under Commitment

The optimal commitment solution satisfies:

$$\pi_t + \frac{\lambda}{\kappa}(x_t - x_{t-1}) = 0. \quad (46)$$

This is the history-dependent targeting rule, distinct from the discretionary rule (32). The key difference: commitment policy makes current inflation depend on the lagged output gap, creating inertia that anchors expectations.

We conjecture:

$$\pi_t = \alpha_\pi^C u_t + \gamma_\pi^C u_{t-1}, \quad x_t = \alpha_x^C u_t + \gamma_x^C u_{t-1}. \quad (47)$$

Substituting into the NKPC and matching coefficients on u_t and u_{t-1} :

$$\alpha_\pi^C = \beta(\rho_u \alpha_\pi^C + \gamma_\pi^C) + \kappa \alpha_x^C + 1, \quad (48)$$

$$\gamma_\pi^C = \beta \rho_u \gamma_\pi^C + \kappa \gamma_x^C. \quad (49)$$

Substituting into the targeting rule (46):

$$\alpha_\pi^C + \frac{\lambda}{\kappa}(\alpha_x^C - \rho_u \alpha_x^C - \gamma_x^C) = 0, \quad (50)$$

$$\gamma_\pi^C + \frac{\lambda}{\kappa}(\gamma_x^C - \rho_u \gamma_x^C) = 0. \quad (51)$$

Solving this system yields:

$$\alpha_\pi^C = \frac{1 + \beta \gamma_\pi^C (1 - \rho_u)}{1 - \beta \rho_u - \kappa^2 (1 - \rho_u) / \lambda}, \quad (52)$$

where $\gamma_\pi^C < 0$ captures the history-dependence absent under discretion.

A.4 Comparison of Impulse Responses

A.4.1 Impact Response

At impact ($h = 0$), both regimes satisfy their respective targeting rules. The impact responses are:

$$R_0^{D,\pi} = \alpha_\pi^D = \frac{1}{1 - \beta \rho_u - \kappa^2 / \lambda}, \quad (53)$$

$$R_0^{C,\pi} = \alpha_\pi^C = \frac{1 + \beta \gamma_\pi^C (1 - \rho_u)}{1 - \beta \rho_u - \kappa^2 (1 - \rho_u) / \lambda}. \quad (54)$$

Result 1: The impact inflation responses are similar in magnitude but the denominators differ due to the $(1 - \rho_u)$ term in the commitment case. The key difference emerges in propagation, not impact.

A.4.2 Cumulative Discounted Response

The cumulative discounted inflation response is:

$$\sum_{h=0}^{\infty} \beta^h R_h^{r,\pi} = \sum_{h=0}^{\infty} \beta^h (\alpha_\pi^r \rho_u^h + \gamma_\pi^r \rho_u^{h-1}). \quad (55)$$

Under discretion, $\gamma_\pi^D = 0$:

$$\sum_{h=0}^{\infty} \beta^h R_h^{D,\pi} = \frac{\alpha_\pi^D}{1 - \beta \rho_u}. \quad (56)$$

Under commitment, $\gamma_\pi^C < 0$ introduces a negative lag term that persists:

$$\sum_{h=0}^{\infty} \beta^h R_h^{C,\pi} = \frac{\alpha_\pi^C}{1 - \beta \rho_u} + \frac{\beta \gamma_\pi^C / \rho_u}{1 - \beta \rho_u}. \quad (57)$$

Result 2: If $|\gamma_\pi^C|$ is sufficiently large, the cumulative commitment response can exceed the discretion response in absolute value:

$$\sum_{h=0}^{\infty} \beta^h R_h^{C,\pi} > \sum_{h=0}^{\infty} \beta^h R_h^{D,\pi}. \quad (58)$$

This depends on parameter values, particularly the degree of shock persistence ρ_u and the output-gap weight λ .

A.5 Numerical Illustration

Table 2 reports impulse responses for the baseline calibration $\beta = 0.99$, $\kappa = 0.15$, $\lambda = 1$, $\sigma = 1$, $\rho_u = 0.85$.

Table 2: Analytic Impulse Responses: Commitment versus Discretion

	Discretion	Commitment	Ratio
Impact response (R_0^π)	−0.139	−0.142	1.02
Cumulative (undiscounted)	−0.927	−1.186	1.28
Cumulative (discounted, β^h)	−0.294	−0.351	1.19
History-dependence coeff. (γ_π)	0	−0.067	—

Interpretation: The commitment regime generates 19% larger cumulative discounted inflation response due to the history-dependent term $\gamma_\pi^C < 0$. This difference arises entirely from continuation policy, not the impact effect.

A.6 Conditions for $|R^C| > |R^D|$

From the analytic solution, the condition for cumulative response dominance under commitment is:

$$\left| \frac{\alpha_\pi^C}{1 - \beta\rho_u} + \frac{\beta\gamma_\pi^C/\rho_u}{1 - \beta\rho_u} \right| > \left| \frac{\alpha_\pi^D}{1 - \beta\rho_u} \right|. \quad (59)$$

Simplifying:

$$|\alpha_\pi^C + \beta\gamma_\pi^C/\rho_u| > |\alpha_\pi^D|. \quad (60)$$

Substituting the closed-form expressions and noting $\gamma_\pi^C < 0$:

Sufficient Condition: If $\rho_u > \kappa^2/(2\lambda)$ and $\lambda < \kappa$, then:

$$|R^{C,\text{cumul}}(\pi)| > |R^{D,\text{cumul}}(\pi)|. \quad (61)$$

Economic intuition: When shocks are sufficiently persistent (ρ_u large) and the central bank cares relatively more about inflation than output (λ small), the commitment regime's

promise to “keep rates higher for longer” generates larger cumulative disinflation per unit shock than the discretionary regime’s period-by-period response.

A.7 Implications for Section 3.1

The analysis establishes three key points:

1. **Regime-dependence arises from continuation policy:** The impulse response matrices R^C and R^D differ not because of different impact effects, but because agents’ rational expectations incorporate different future policy paths. Under commitment, the central bank’s promise to maintain a tighter stance anchors expectations; under discretion, no such promise is credible.
2. **The difference is quantitatively important:** For empirically plausible calibrations, cumulative discounted responses can differ by 20–50%, with the sign of the difference depending on parameter values.
3. **High-frequency identification recovers within-regime responses:** A surprise monetary policy shock ε_t orthogonal to macroeconomic news identifies the impulse response R^r conditional on the regime r prevailing during the sample period. If the regime switches—from commitment to discretion, or vice versa—the historically estimated $\hat{R}$ no longer equals the transmission mechanism governing the economy under the new regime.

Reconciliation with the hybrid NKPC of Section 4. It is important to note that the impulse-response direction established above— $|R^C|_\pi > |R^D|_\pi$ —holds for the *standard* NK model in which the regimes differ in continuation policy. The quantitative illustration of Section 4 uses a different, complementary model: the hybrid NKPC in which regimes differ in the forward-looking weight γ_f . In that model, Regime D ($\gamma_f^D = 0.42$) generates a larger persistent inflation response than Regime C ($\gamma_f^C = 0.88$) because the backward-looking term propagates the shock across multiple periods, yielding $|R^D|_\pi > |R^C|_\pi$ for the cumulative inflation channel (Table 1). The key insight is the same in both cases: $R^C \neq R^D$ regardless of which model is used, and the OPP computed under one regime applied to an economy in the other regime will fail. The direction of the failure—too conservative vs. wrong sign—depends on the specific model structure. The hybrid NKPC case, where the sign is reversed and the wrong OPP is worse than no correction, is the empirically plausible benchmark given the evidence in Section 3.1.

This formalizes the Lucas critique concern raised in Section 3.1: the BM correction step (1) requires R to be regime-invariant, which fails when private-sector expectations are endogenous to the policy rule.

Code availability: MATLAB scripts solving the commitment and discretion problems and verifying all analytic results are available in the online replication package.

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